Introduction to Mobile Robotics

CHAPTER 1 : OVERVIEW

Lectures	Tutorials	Labs
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ENP	CDTA	CDTA

SUMMARY

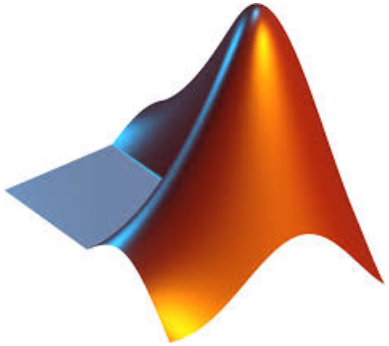
- Ch1. Mobile Robotics overview
- Ch2 : Homogenous transformations & Orientation Representations
- Ch3 : Dynamic Modeling and Control Theory
-

- ***----------MID TERM EXAM ----------***
- Ch4 : Kinematics & Dynamics
- Ch5 : Localization & Mapping
- Ch6 : Motion control , Path Planning and Obstacle Avoidance

- ***----------FINAL EXAM ----------***

SIMULATION TOOLS

- MATLAB / SIMULINK



- PYTHON



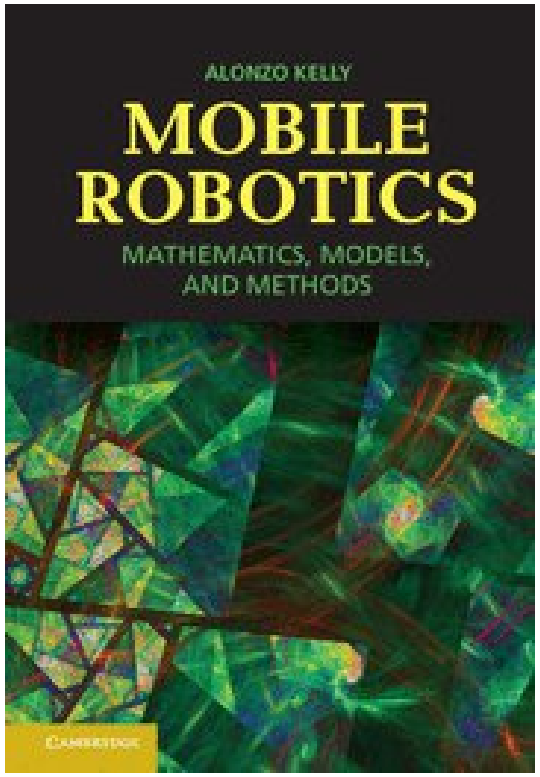
- ROS

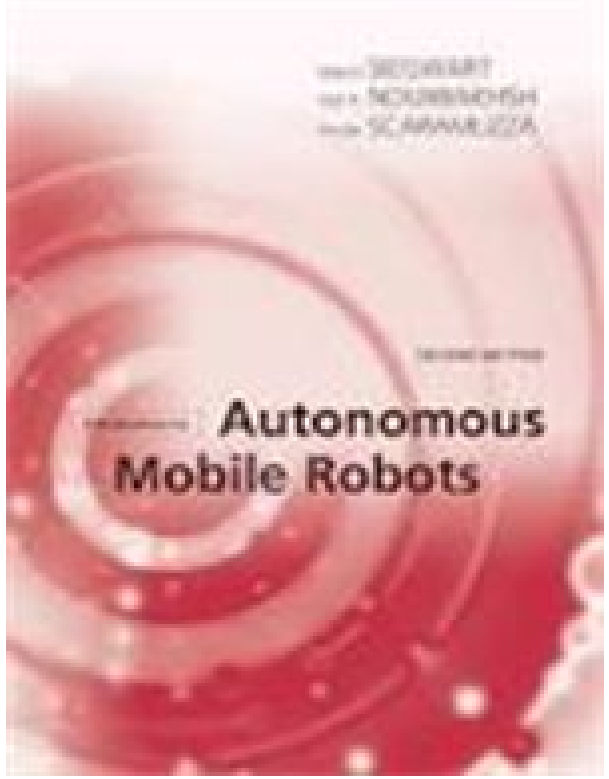


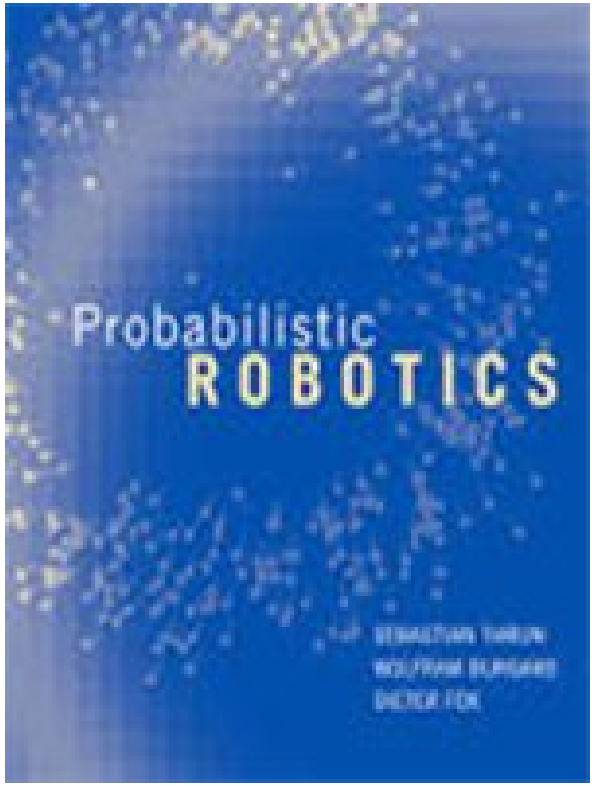
- VREP

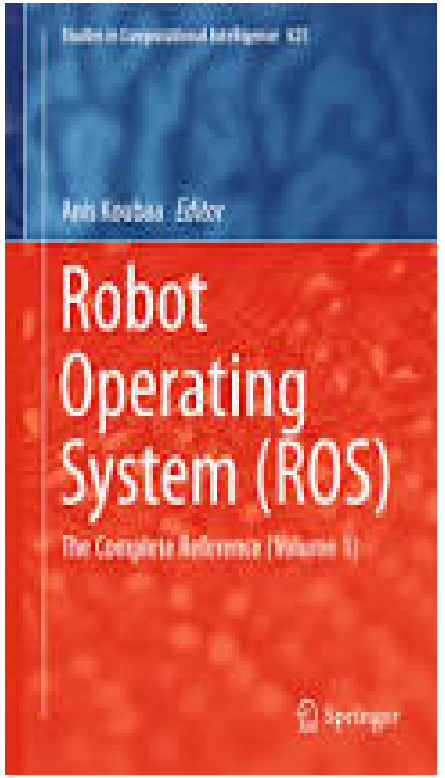


REFERENCES


ALONZO KELLY
MOBILE ROBOTICS
MATHEMATICS, MODELS, AND METHODS


Autonomous Mobile Robots


Probabilistic ROBOTICS


Robot Operating System (ROS)

HISTORY

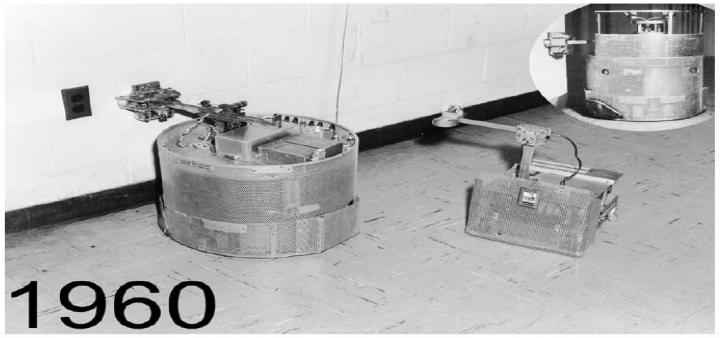
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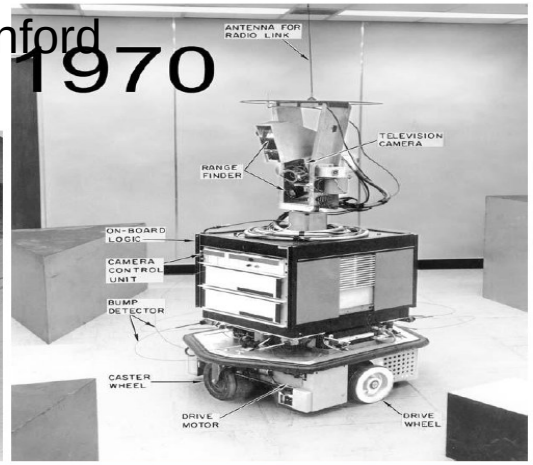
•The “word” robot was first introduced in 1920, by the writer Karel Kaspeck in his play named “Rossum’s Universal Robots”. The term gets its origin from Slavic roots associating it to “Labor”.

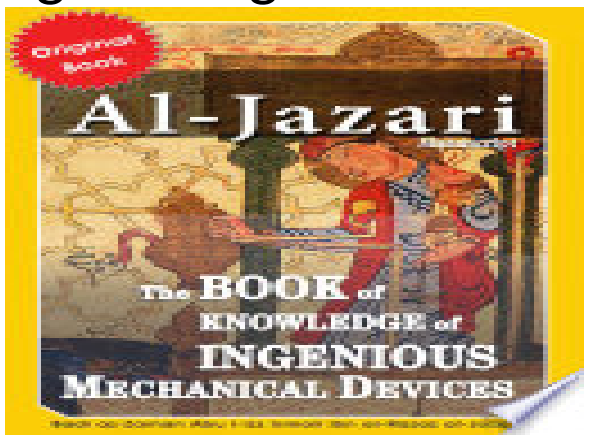
• Ismail al-jazari wrote the “Book of Knowledge of Ingenious Mechanical Devices” (Kitab fi maʿrifat al-hiyal al-handasiyya.) where he described 50 mechanical devices along with instructions on creating them. He has been described as the father of robotics and modern day engineering.

•)

•Beast (John Hopkins University) & Shakey (Stanford University)







DEFENITIONS

1- Robot Manipulator : A reprogrammable, multifunctional machine designed to act on material, parts, tools or specialised devices through various programmed motions for the performance of a variety of tasks

2- Mobile Robot :A machine equipped with perception, decision and action capabilities that allow it to act autonomously on its environment







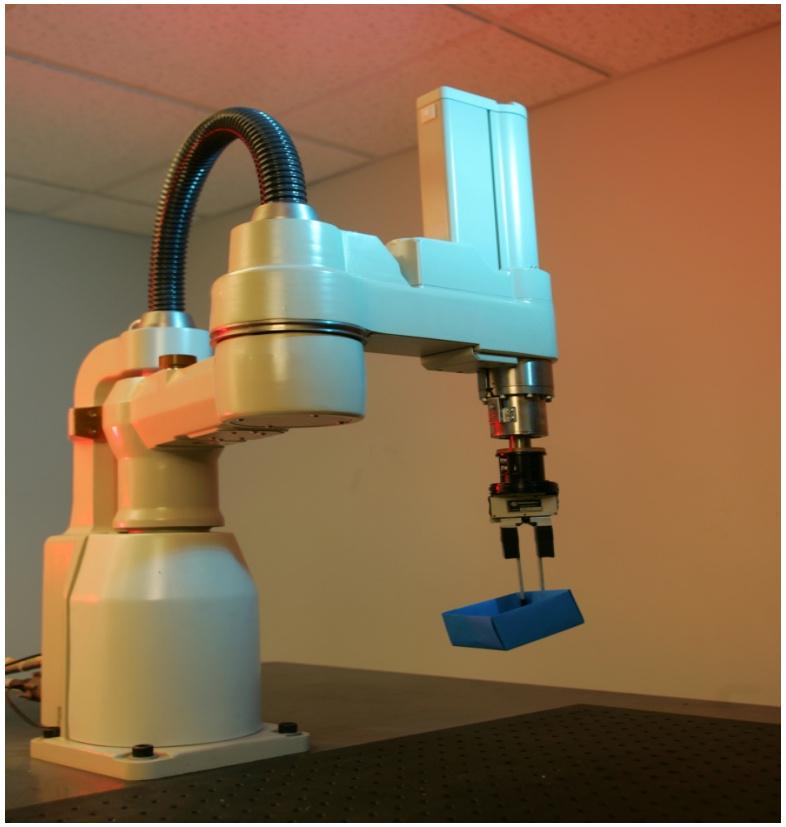
Mobile Robotics: Multi-Disciplinary Domain

- Mobile Robotics is a multi-disciplinary field. Best robotics researchers and engineers will touch upon all disciplines:

- **Mechanical Engineering** – concerned primarily with manipulator/mobile robot design, kinematics, dynamics, compliance and actuation.
- **Electrical & Control Engineering** – concerned primarily with robot actuation, electronic interfacing to computers and sensors, and control algorithms.
- **Computer Science & AI** – concerned primarily with robot programming, planning, decision making and intelligent behavior.

Manipulators Examples





Puma (Programmable Universal Machine for Assembly), by Unimation,

SCARA (Selective Compliant Articulated Robot for Assembly) introduced in Japan and the US (by Adept Technologies), .

Mobile Robots Examples



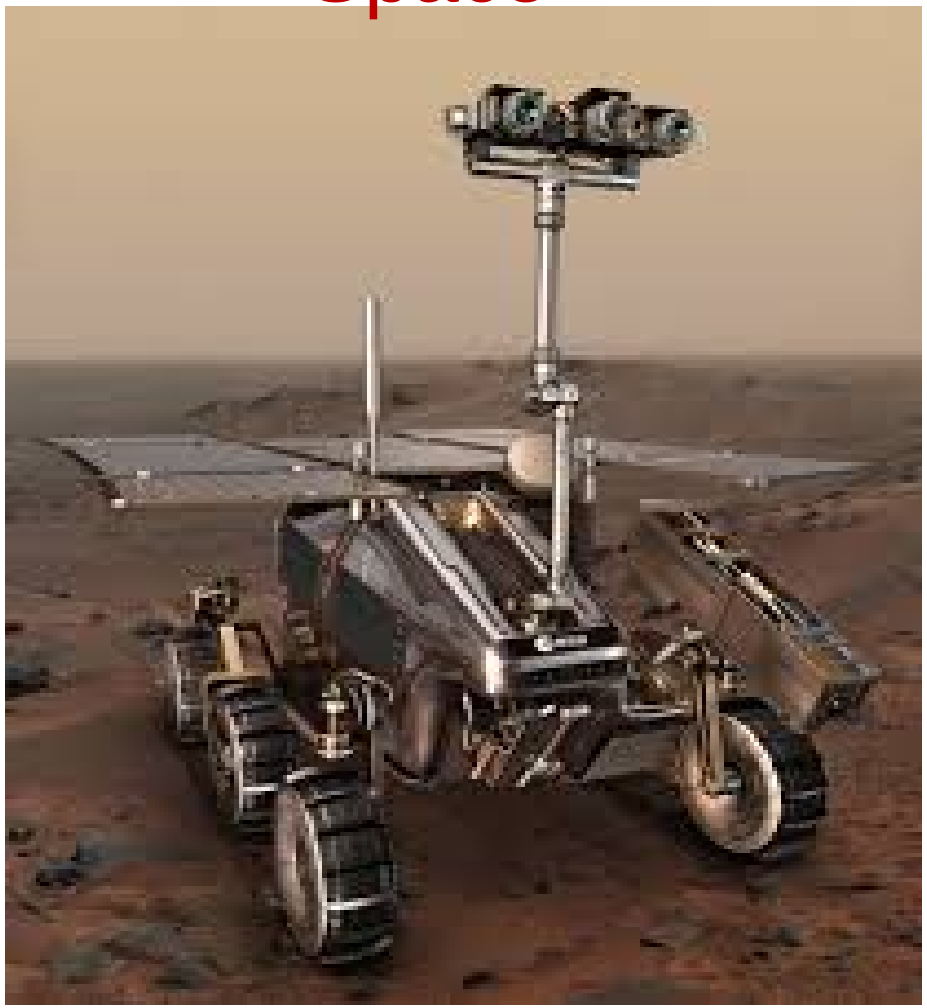
From Boston Dynamics



Legged Robots

Wheeled Robots

Space



Mars Rovers

Under Water



Aerial Mobile Robots



Skydio: 'The Self-Flying Camera'



- Visual-inertial navigation and obstacle avoidance using multiple stereo camera pairs.
- Mobile NVidia processor onboard.
- `https://www.youtube.com/watch?v=gsfkGlSajHQ`

Drones

WALKING ROBOT



- Spot, from Boston Dynamics `https://youtu.be/wlkCQXHEgjA`.
- Five active stereo sensors give all around perception for localisation and mapping.
- Product officially launched in 2019 and aimed at industrial inspection applications.

Other types of Robots….



From forbes.com

Autonomous Cars



Honda ASIMO, the first non-prototype release of its humanoid robot, .



From ESA.int



From ESA.int

Autonomous Space Vehicles

https://blogs.esa.int/cleanspace/2018/06/01/esa-opens-the-renegade-activity-for-ssv/

Examples

Fully Automated BMW Car Factory
https://www.youtube.com/watch?v=CBmsOvbGh-Y

Amazon warehouse robots
Amazon Warehouse Order Picking Robots - YouTube

More Advanced ones
https://www.youtube.com/watch?v=fn3KWM1kuAw

Mobile Robots VS Manipulator Arms

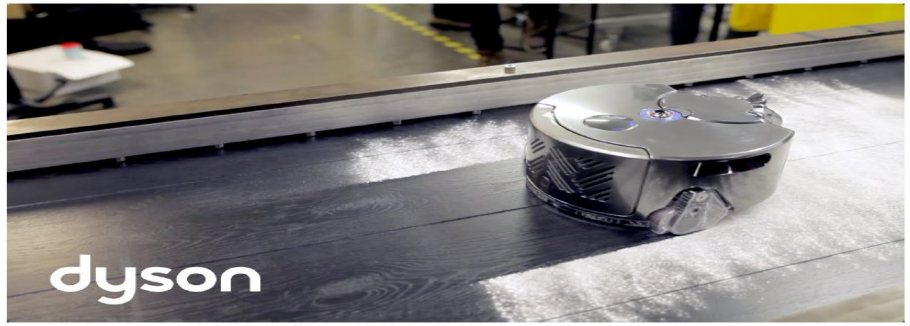
Strong & precise articulated arms to accomplish tasks that were performed by humans – “**articulated robots”, or “manipulators”. Fear that they will replace human laborers.**

Use of mobility to reposition the robot from one location to another, “**mobile robots”. This can be done by locomotion like humans do (“legged robots”), but most likely it will use other means such as wheels (“wheeled robots”).**

Mobile Robots VS Manipulator Arms

- Both platforms need to calculate their kinematics
- The working space operation for the mobile robots is unbounded comparing to the one of the fixed manipulator
- The position of the mobile robot is provided by external sensors leading to higher inaccuracies.
- Mobile robots can be built in different structures to adapt to their environment of operations. They could be as wheeled robots, drones as aerial robots, humanoid robots...etc



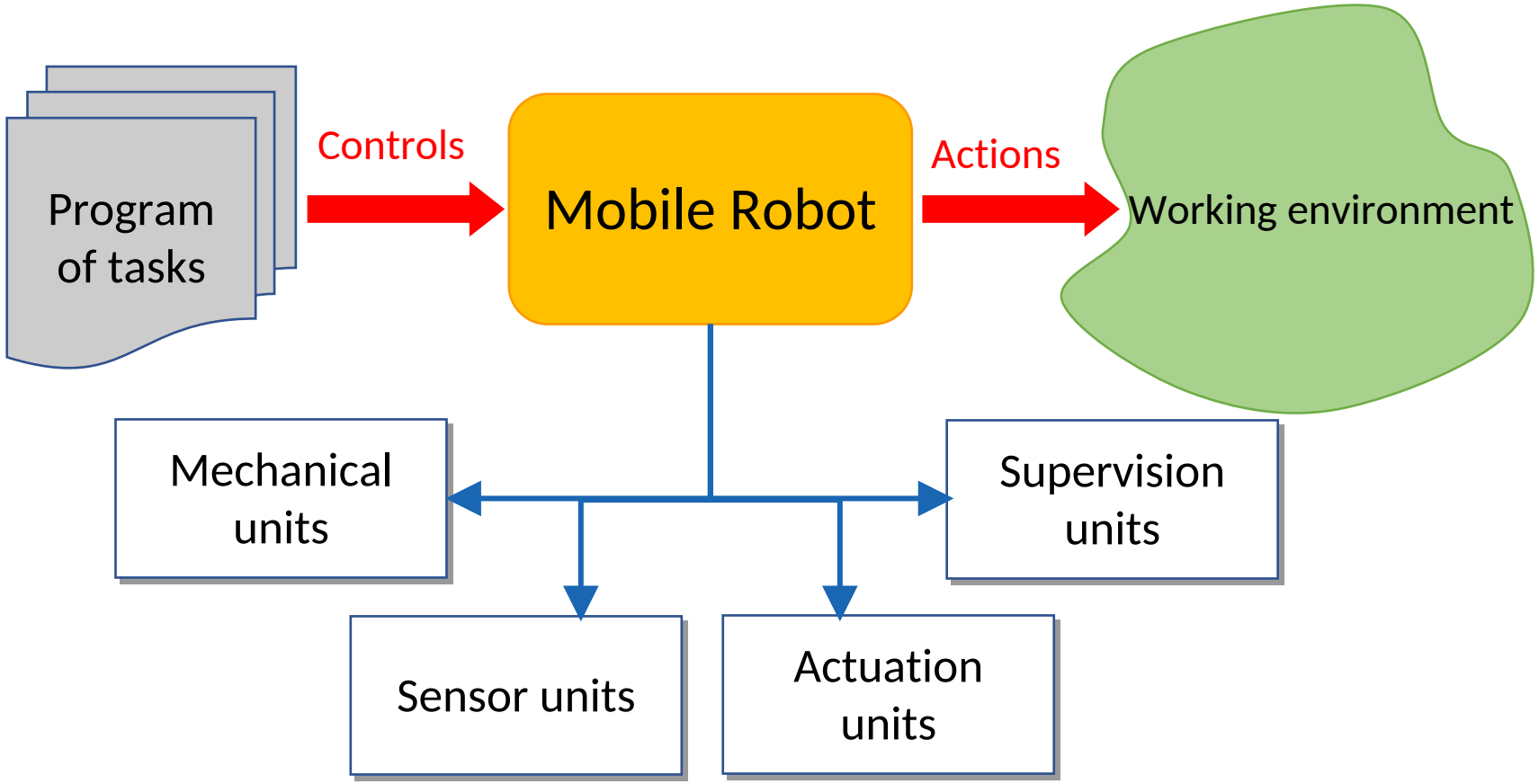


- On sale in Japan in 2015; around the world in 2016.
- Uses omnidirectional vision to build a map of its environment automatically (SLAM). This permits accurate, repeatable localisation, and therefore precise coverage and cleaning.
- https://youtu.be/gZ3VV001Axo

Classification of Mobile Robots

- Way of movement : wheeled mobile robots, walking mobile robots (single-leg, double-leg and multi-leg), Aerial mobile robots, crawling robots, creeping robots and swimming robots and other types.
- Working environment : indoor mobile robots and outdoor mobile robots.
- Functions and uses : medical robots, military robots, disabled robots, cleaning robots, etc.

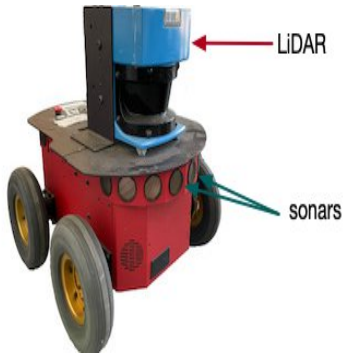
Mobile Robot Structure

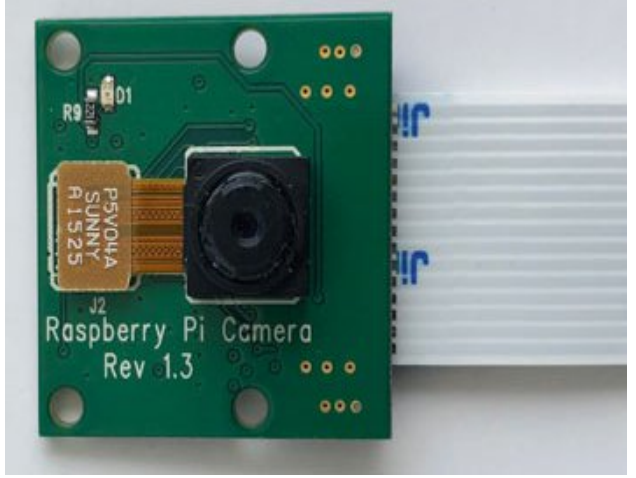


SENSOR UNITS









- A sensor is an instrument measuring a physical property of the Mobile Robot or its environment
- Sensors are noisy, imprecise and limited;
- The environment of a robot is generally complex, changing, unpredictable and uncertain

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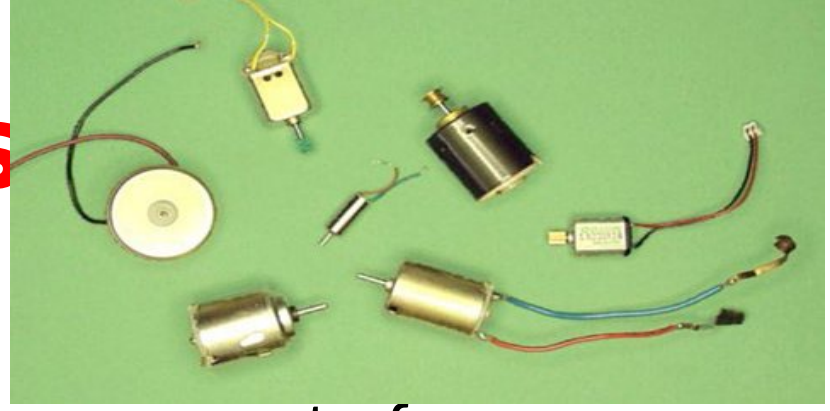
- **Proprioceptive sensors** for monitoring internal robot state (e.g. position and velocity, torque, acceleration, orientation).

- **Exteroceptive sensors** to obtain information from the robot environment (e.g distance to objects, obstacles, land marks).

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Actuating units



- An actuator (electric, pneumatic, hydraulic) is a component of a machine that is responsible for moving a the mobile robot;
- An actuator controls a degree of freedom (rotation, translation);
- Actuators could be complex



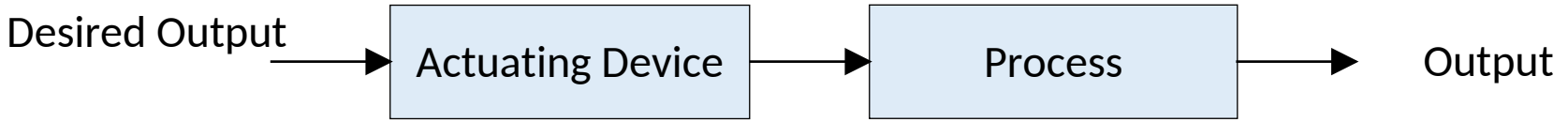




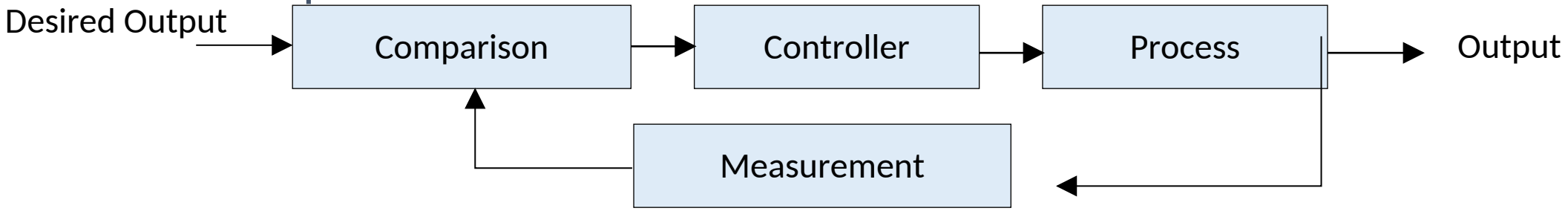


Control Unit

- **Open loop control system:** Uses an actuating device to control the process directly without using feedback.



- **Closed-loop control system:** Uses a measurement feedback of the output signal to compare the output with the desired output or reference.



MOTION FREEDOM

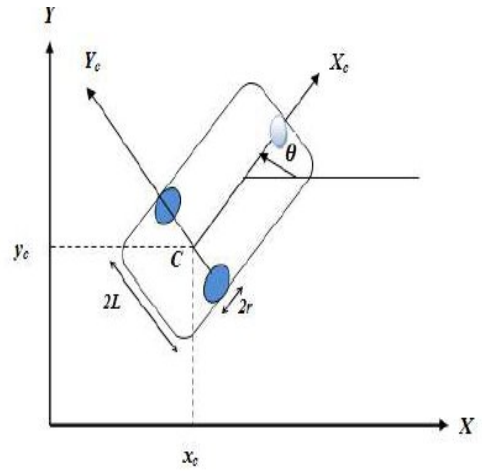
- A rigid body which translates and rotates along a 1D path has 1 degree of freedom (DOF): translational. Example: a train.
- A rigid body which translates and rotates on a 2D plane has 3 DOF: 2 translational, 1 rotational. Example: a ground robot.
- A rigid body which translates and rotates in a 3D volume has 6 DOF: 3 translational, 3 rotational. Example: a flying robot.
- A *holonomic robot* is one which is able to move instantaneously in any direction in the space of its degrees of freedom.
- Otherwise a robot is called *non-holonomic*.

ROBOT POSE

- If we assume that a robot is confined to moving on a plane, its location can be defined with a *state vector* **x** consisting of three parameters:

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix}$$

- x and y specify the location of the pre-defined 'robot centre' point in the world frame.
- θ specifies the rotation angle between the two coordinate frames (the angle between the x^W and x^R axes).
- The two coordinate frame coincide when the robot is at the origin, and $x = y = \theta = 0$.



Standard Wheel Configurations



Drive and Steer

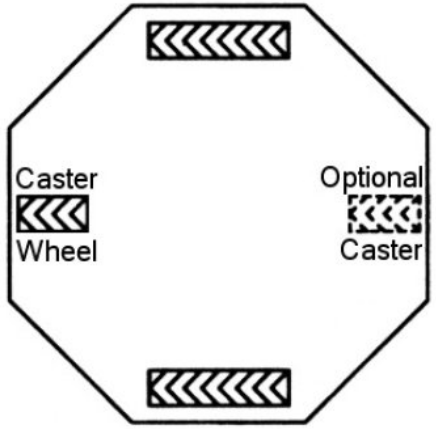


Differential Drive

- Simple, reliable, robust mechanisms suitable for robots which essentially move in a plane.
- Both of these configurations are *non-holonomic* (each uses two motors, but has three degrees of movement freedom). For instance, a car-like robot can't instantaneously move sideways.

Differential Drive

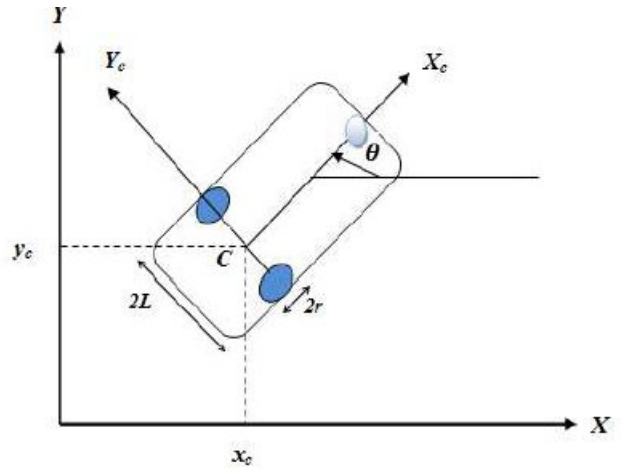




- Two driving wheels on the left and right sides of a robot, each driven by its own motor. Steering is achieved by setting different wheel speeds.
- Wheels run at equal speeds for straight-line motion.
- Wheels run at equal and opposite speeds to turn on the spot.
- Other combinations of speeds lead to motion in a circular arc.

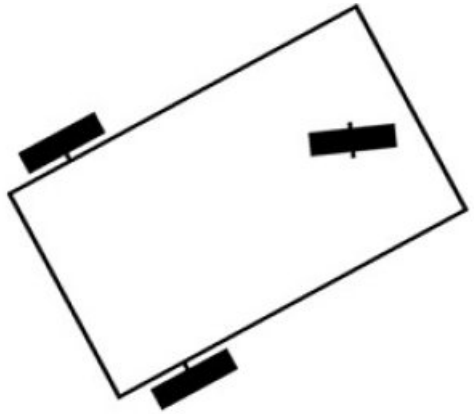
UNICYCLE MOBILE ROBOTS (DD)

- Operated by two independent motorized wheels
- Has possible steering free wheel to ensure its stability
- Under actuated system, simple control



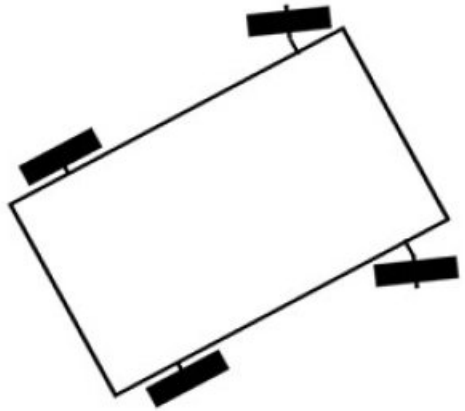
TRICYCLE MOBILE ROBOTS (DS)

- Two fixed wheels placed on the same axis + one centred orientable wheel (longitudinal axis)
- Movement given by the speed V of the fixed wheels and the angular velocity w of the orientable wheel
- Under actuated system, complex control



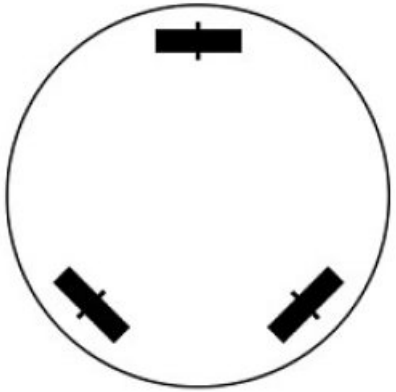
CAR LIKE MOBILE ROBOTS (DS)

- Two fixed wheels placed on the same axis + Two centred orientable wheel placed on the same axis
- More Stable than Tricycle
- Under actuated system, complex control



OMNIDIRECTIONAL MOBILE ROBOTS

- Three motorized wheels arranged in an equilateral triangle
- Can move freely in all directions
- Simple Control



2D/3D Rotations

- 2D :

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

- 3D :

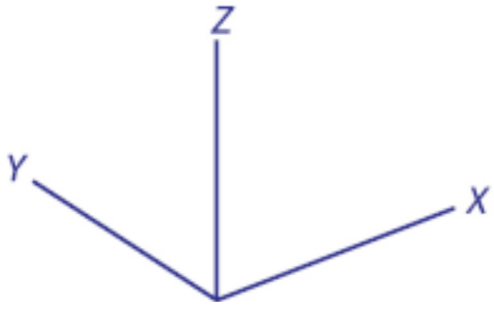
$$R_{\hat{x}}(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \cos\phi & \sin\phi \end{pmatrix} \qquad R_{\hat{y}}(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \qquad R_{\hat{z}}(\psi) = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\begin{aligned} R &= R_{\hat{x}}(\phi) R_{\hat{y}}(\theta) R_{\hat{z}}(\psi) \\ &= \begin{pmatrix} c(\theta)c(\psi) & s(\phi)s(\theta)c(\psi) - c(\phi)s(\psi) & c(\phi)s(\theta)c(\psi) + s(\phi)s(\psi) \\ c(\theta)s(\psi) & s(\phi)s(\theta)s(\psi) + c(\phi)c(\psi) & c(\phi)s(\theta)s(\psi) - s(\phi)c(\psi) \\ -s(\theta) & s(\phi)c(\theta) & c(\phi)c(\theta) \end{pmatrix} \end{aligned}$$

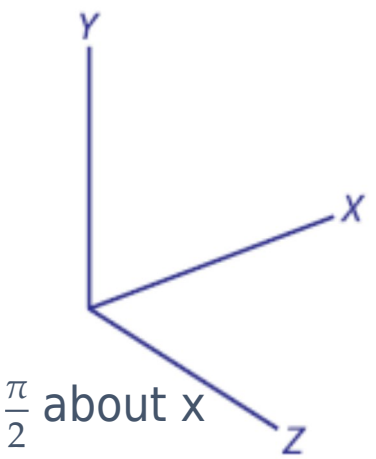
Rotations

- **Rotation matrix properties**
  - $R^T = R^{-1}$ or $RR^T = R^T R = I$
  - $\det(R) = \pm 1$
  - Chain rule property: $R_i^k = R_j^k \, R_i^j$

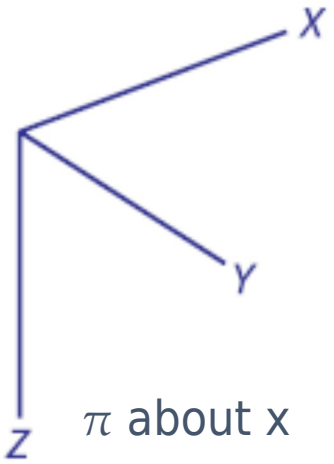
- **Examples:** *Identify the following rotations*



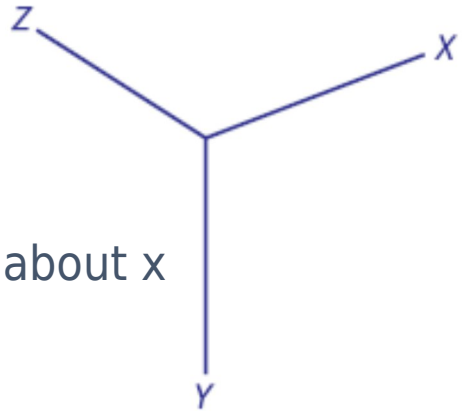
Original



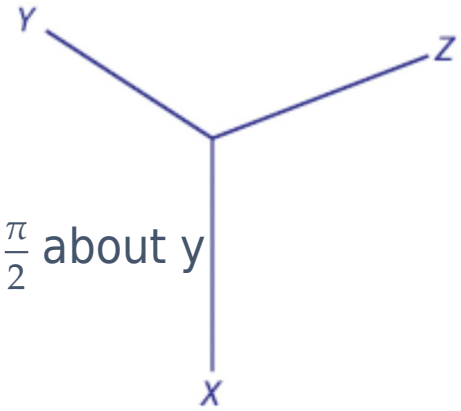
$\frac{\pi}{2}$ about x



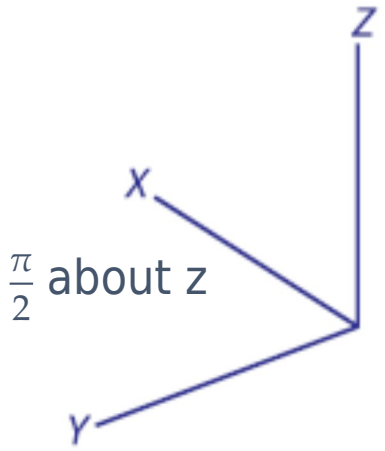
π about x



$-\frac{\pi}{2}$ about x



$\frac{\pi}{2}$ about y



$\frac{\pi}{2}$ about z

Homogenous transformations

- We can consider position (translation) and orientation (rotation) separately.
  - **Translation** is performed through vector addition.
  - **Rotation** is performed through matrix-vector, matrix-matrix multiplication.
- It is suitable to have a uniform representation of a translation, rotation or both with the same matrix.
- This is achieved by **homogenous transformations**.

Homogenous transformations

- **Translation**
  - Consider applying a translation $\boldsymbol{d}$ to a point $\boldsymbol{p}$.

$$p + d = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} + \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix} = \begin{bmatrix} p_x + d_x \\ p_y + d_y \\ p_z + d_z \end{bmatrix}$$

  - We want to convert this vector addition into a matrix-vector multiplication:

$$\begin{bmatrix} \quad ? \quad \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} p_x + d_x \\ p_y + d_y \\ p_z + d_z \end{bmatrix}$$

Is this possible?

Not possible !
Multiplying a matrix by a vector gives a matrix with the same rank.

Homogenous transformations

- Let us consider the following augmented forms of vector p_h and matrix D

$$p_h = \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} \qquad D$$

$$D \times p_h = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = \begin{bmatrix} p_x + d_x \\ p_y + d_y \\ p_z + d_z \\ 1 \end{bmatrix}$$

Problem solved !

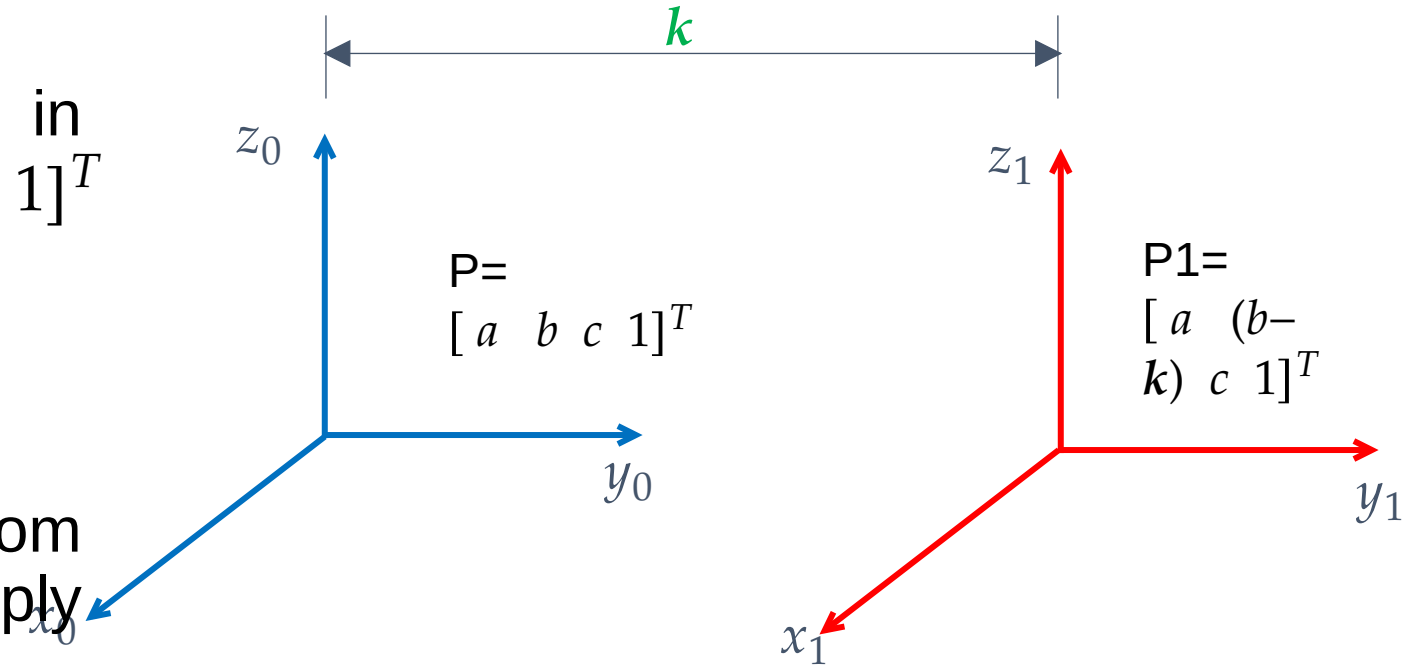
Through augmented matrix.

We converted this vector addition into a matrix-vector multiplication

Homogenous transformations

- Let us consider the following two frames {0} {1}. Frame {1} is translated from frame {0} by k units along the y-axis.

- A homogeneous point $[\,a \quad b \quad c \quad 1]^T$ in frame {0} is the same as $[\,a \quad (b–\boldsymbol{k}) \quad c \quad 1]^T$ in frame {1}.



- Another way, to convert points from frame {0} to frame {1} is to pre-multiply the point by the matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\boldsymbol{k} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogenous transformations

- **Examples:**

Location in {0}

$$\begin{bmatrix}0\\0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\-3\\0\\1\end{bmatrix}$$

Location in {1}

$$\begin{bmatrix}0\\-3\\0\\1\end{bmatrix}$$

Using pre-multiplication

$$\begin{bmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & -3\\0 & 0 & 1 & 0\\0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0\\0\\0\\1\end{bmatrix} =$$

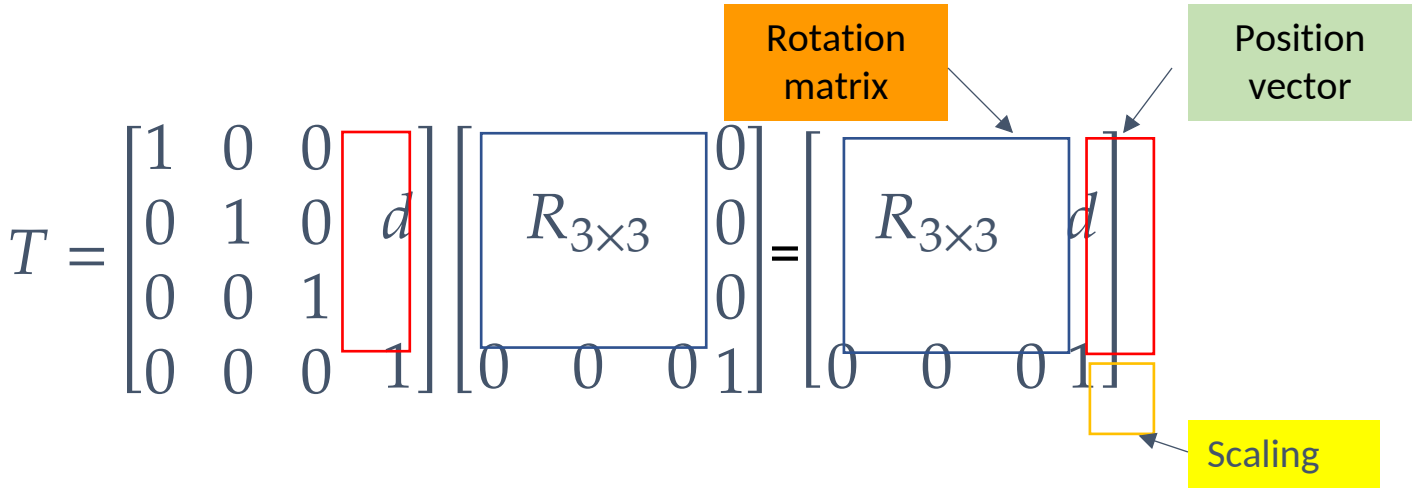
$$\begin{bmatrix}0\\3\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}0\\0\\0\\1\end{bmatrix}$$

$$\begin{bmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & -3\\0 & 0 & 1 & 0\\0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0\\3\\0\\1\end{bmatrix} =$$

Homogenous transformations

- The two alternatives presented in the previous slides lead to the same transformation.



T is 4x4 transformation matrix

R is 3x3 rotation matrix

d is 3x1 translation matrix

Linear Algebra : Fundamentals

- Matrix Product : A*B # B*A
- Rank
- Inverse & Pseudo Inverse
- Eigenvalues & Singular Values
- Condition Number : Cond(A) =Sigma_max(A)/Sigma_min(A)

European Ariane 5 Launcher explode in flight (June 4 1996). Faillure caused by computer malfunction (CN >> 1) !!!

https://youtu.be/wGeZEUjUKvc

https://www.youtube.com/watch?v=wGeZEUjUKvc

Condition Number

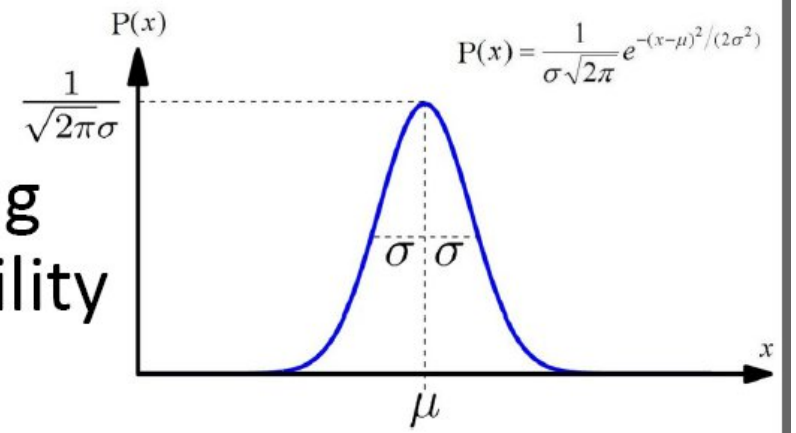
- Cond(A) =Sigma_max(A)/Sigma_min(A)

- System of equation A*x=b x=inv(A)*b
- Well conditioned : small changes in A elements small changes in x
- ill conditioned : small changes in A elements big changes in x

- Cond(A) around 1 : well conditioned matrix
- Cond(A) >10 :ill conditioned matrix

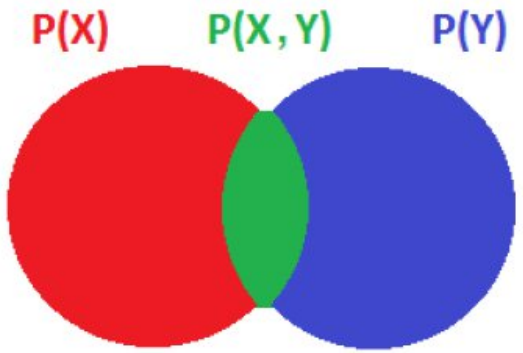
PROBABILITY RULES

- X represetns a random variable
- X can take countable number of values: $\{x_1, x_2, .., x_n\}$
- $P(\cdot)$ represents the probability function e.g. Gaussian, Uniform etc.
- $P(X = x_i)$ or $P(x_i)$ is the probability of occurring event x_i using the probability function $P(\cdot)$



RECAP OF PROBABILITY

- $0 \leq P(X) \leq 1$
- $P(TRUE) = 1$
- $P(FALSE) = 0$
- $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$
- $P(\neg X) = 1 - P(X)$
- $\sum_x P(x_i) = 1$



PROBABILITY RULES

- Joint Probability
  - $P(X = x \; and \; Y = y) = P(x \cap y) = P(x, y)$
  - If X and Y are independent $P(x, y) = P(x) \cdot P(y)$

- Conditional Probability
  - $P(x|y) = \frac{P(x,y)}{P(y)}$ or $P(x, y) = P(x|y) \cdot P(y)$
  - If X and Y are independent $P(x|y) = P(x)$

PROBABILITY RULES

- Law of Total Probability
  - $P(y) = \sum_x P(y|x) \cdot P(x)$
- Marginalization
  - $P(x) = \sum_y P(x, y)$
- Bayes Formula
  - $P(x, y) = P(x|y) \cdot P(y) = P(y|x) \cdot P(x)$

$$\Rightarrow P(x|y) = \frac{P(y|x) \cdot P(x)}{P(y)} = \frac{expected \cdot piror}{measurement}$$